

Thalassemia Diagnosis Through Medical Imaging: A New Artificial Intelligence-Based Framework

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Abstract—Thalassemia is the most common single-gene disorder throughout the world and represents a major public health problem resulting in abnormal ratios of hemoglobin subunits. It is widely spread throughout the Mediterranean region, Middle East, Southeast Asia, and some parts of Africa. There are different types of Thalassemia characterized by abnormal hemoglobin production. Most of the current hemoglobinopathy screening methods include High Performance Liquid Chromatography (HPLC), hemoglobin electrophoresis, screening of Polymerase Chain Reaction (PCR) mutations, and Deoxyribonucleic Acid (DNA) tests. However, all these methods are costly and require specialized instrumentation and trained technicians. Despite some studies used Artificial Intelligence (AI) they focused merely on Machine Learning (ML) and results were not optimal. This study aims to design a new AI-based framework for Thalassemia diagnosis using Deep Learning (DL) and a new combination of metrics for evaluation. This was achieved through the development and evaluation of a supervised semantic image segmentation model, and the implementation of different data engineering methods such as data annotation, augmentation, pre-processing, and preparation. Transfer learning was utilized and the Prediction Time Augmentation (PTA) was employed to get smoother and more accurate predictions. Quantitative results showed that, the mean Intersection Over Union (IoU) score of prediction of Thalassemia was 88% with PTA and 82% without PTA. Results also showed that as the combined metric of loss scores decreases the prediction of Thalassemia increases. Qualitative results showed that the final prediction of Thalassemia focuses on the Codocytes and labels the other unidentified cells as the background. Also, the resulting image was smoother and less bulky than the original annotated ground truth, and thus could be feasibly diagnosed. As a future prospect we aim to implement more algorithms and extend the diagnosis to include other diseases.

Keywords—Thalassemia; Codocyte; Prediction time augmentation; Combined Metric Loss; Intersection over Union; Medical Image Processing; Artificial Intelligence; Deep learning.

I. INTRODUCTION

Thalassemia is a genetic blood disorder that causes the body to have reduced hemoglobin levels as compared to the normal level. Thalassemia is caused by missing or damaged genes that are responsible for hemoglobin production. Hemoglobin (Hb) is the protein that enables red blood cells,

also known as erythrocytes, to carry oxygen. Thalassemia is common mostly in African Americans and people of Asian Southeast or Mediterranean descent. Thalassemia is divided into two main types, alpha and beta, depending on the affected genes. Symptoms range from none, to having varied severity levels of anemia, and in extreme cases, death before birth. Symptoms can appear at early stages in life and may affect the patient's life span. It has no cure, but treatment can reduce the symptoms [1-3].

Many tests were applied to diagnose Thalassemia, some of which include Complete Blood Count (CBC), Hemoglobin electrophoresis with A2 and F-quantitation, Free Erythrocyte Protoporphyrin (FEP) and ferritin, Chorionic Villus Sampling (CVS), as well as Amniocentesis [3]. However, these methods were either expensive, time consuming, or prone to human error. Moreover, such reported tests were usually done in clinics and hospitals, merely.

Artificial Intelligence (AI) and Machine Learning (ML) algorithms associated with Medical Imaging were employed to resolve the aforementioned problems and improve the clinical diagnosis. AI was used for Thalassemia detection by Chakraborty et al., who developed a computer vision approach for Thalassemia screening on Scanning Electron Microscopic (SEM) images [4]. The study utilized geometric features that characterizes the erythrocyte's morphology. Erythrocytes were separated from blood samples and then chemically processed for imaging at 2000× resolution [4]. Next, marker-controlled watershed transform was used to segregate erythrocytes from the SEM images. Geometric features were extracted, and few were found to be statistically significant in discriminating Thalassemia and normal erythrocytes. Finally, a Multilayer Perceptron (MLP) neural network was trained and tested using these features. According to the study, MLP provided a high overall accuracy of 94.59% with 90.38% sensitivity, 98.81% specificity and 98.93% positive predictive value [4]. Another research done by Yousefian et al. utilized AI with the complete blood count (CBC) test parameters as features to diagnose and predict Thalassemia [5]. Different ML architectures were trained and evaluated such as, k-Nearest Neighbors' algorithm (KNN), Multilayer Perceptron (MLP), Neural Network Decision Tree (NN DT) and SVM. As a result, the most effective factors to diagnose thalassemia are Red Blood Count (RBC), Hemoglobin (HGB), Mean Corpuscular Volume (MCV), and Hematocrit (HTC). The accuracy, in percentage, of each method with the respectively used parameters are shown in Fig. 1 [5]. The highest accuracy was associated with MLP.

Author name	Feature													Class	Method	Accuracy
	RBC	HGB	MCV	MCH	MCHC	RDW	PLT	NEUT	LUC	WBC	HTC	HT	Age			
Amendolia et.al.														Alpha-thalassemia Beta thalassemia healthy	KNN MLP SVM	93.2%
Dr. Payandeh et.al.														Anemia Thalassemia (ITP) Cancer Blood lymphoproliferative	MLP	99.12%
Elshami and Alhalees														Major	NB	94.35%
														Minor	DT	93.64%
														Intermedia	NN	95.71%
														Minor - Iron Deficiency	NB	94.32%
														iron-Deficiency	DT	93.65%
														Healthy and Etc.	NN	95.48%

Figure 1: Evaluation of the existing methods for diagnosis of Thalassemia [5].

Although the existing AI and ML based studies were promising, yet they did not focus on the use of Deep Learning (DL) associated with medical imaging. Despite AI and DL was recently used by Zaylaa et al., and the mixed algorithm Convolution Neural Network and Support Vector Machine (CNN-SVM) surpassed the performance of the feed forward backpropagation NN, it was associated with optical imaging in order to automatically detect eye diseases [6] rather than Thalassemia.

Various research studies reported the use of automated AI techniques to predict Thalassemia from medical images or differentiate blood diseases from other hematology diseases, however they relied on ML [4,5], and results were not optimal, were not applied directly on Codocyte images to diagnose Thalassemia and were computationally costly. The aim of the study is to provide a more accessible, automated, cheap, accurate, quick framework for Thalassemia detection based on AI and DL in order to improve the diagnosis, and thus improve healthcare services.

The paper is divided into 7 sections, starting with the introduction which includes a review on the existing studies and algorithms, then the new AI-based framework of Thalassemia diagnosis, the design of the new framework and training, the metrics and loss functions, the results, discussion and conclusion, and finally the future work.

II. NEW AI-BASED FRAMEWORK

The study provided a deep learning approach to localize and identify an erythrocyte morphology that aides in the diagnosis of Thalassemia. The new framework employed the use of the U-net architecture [7] for the first time to perform our semantic image segmentation. Moreover, the Transfer

learning [8] techniques were used to initialize the weights of the U-net model using the trained Auto Encoder weights and gain faster convergence. In addition to data augmentation in order to make up for the lack of data and achieve better generalization. Besides, Prediction Time Augmentation (PTA) [9] was employed to get smoother and more accurate predictions. The starting point was the data collection and preparation.

A. Data Collection and Preparation

The data collection and preparation was a crucial step in the AI-based framework of Thalassemia diagnosis, it comprised the image specifications and processing and specifying the Region of Interest (ROI).

A.1 Image Specifications and Processing

The dataset initially consists of a few high-quality resolution blood smear images, each containing target cells as shown in Fig. 2. The images were collected from different sources on the internet [10], and saved as Portable Network Graphic (PNG) image file, because it uses lossless compression which maintains all the data contained in the image. Each image collected was cropped into smaller patches of images with a target cell at the center of each image. The images were resized to 256x256 pixels and has an aspect ratio of 1.

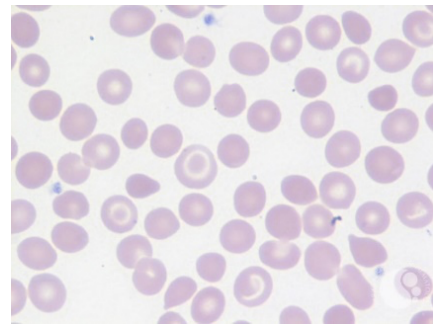


Figure 2: A collected blood smear [11].

A.2 The Region Of Interest (ROI)

Target cells also named as Codocytes, have an excess of cell membrane relative to cell volume. Microcytic target cells appear abundantly in Thalassemia, they are the result of an increased surface-to-volume ratio and signify either excessive loss of internal content of the erythrocyte cell or excessive cell membrane formation. The major causes of decreased Hb production, and, thus, decreased internal volume, include Iron Deficiency Anemia (IDA) and Thalassemia. Hence, the ROI was basically achieved by semantic image segmentation of blood smear images to detect and diagnose Thalassemia.

B. Data Annotation

A supervised binary semantic image segmentation technique was used to detect the ROI. The model took an image as an input and generated an output prediction, such that the ROI was colored in white whereas the background was colored in black. For this technique to work, classification had to happen at a pixelated level targeting certain regions in the image. It must be coupled with a threshold image mask that acts as ground truth for AI to learn from. An image polygonal annotation tool [12] was used to label the boundaries of the

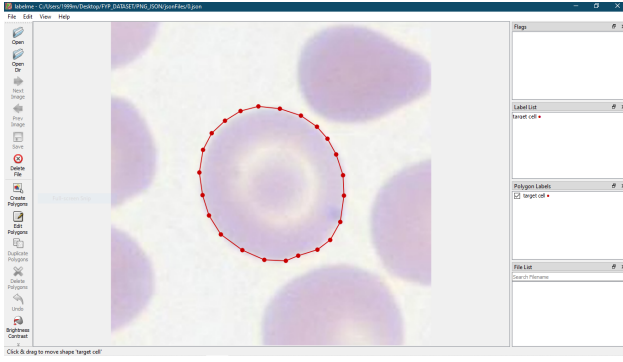


Figure 3: The New AI-Based Framework Result of Polygonal Annotation of Image Named A.

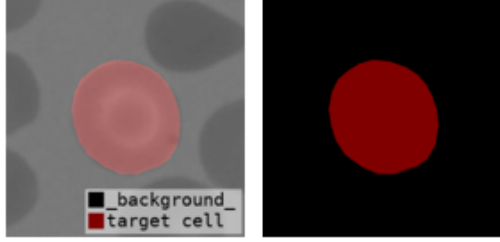


Figure 4: Data Annotation Result of Image A. (Left) Labeled Image. (Right) The Respective Threshold mask.

target cell on each image named ‘A’ as shown in Fig. 3, resulting in a dataset of approximately 500 images and their respective threshold masks as shown in Figs. 4 (a) and (b), respectively.

C. Data Augmentation

Data augmentation was implemented every few epochs to make up for the lack of data and avoid overfitting [13]. Different techniques were applied, such as affine transformations, noise, and filters with tweaked and tuned parameters in a way that retains the learnable features of the images and maintain high information instead of data loss and data leakage [14]. The applied techniques had random probabilities for each of them to occur. The original and multiple augmentations are shown in Fig. 5 starting from the top left original image to the top center to bottom left consecutive multiple augmentations on the original image.

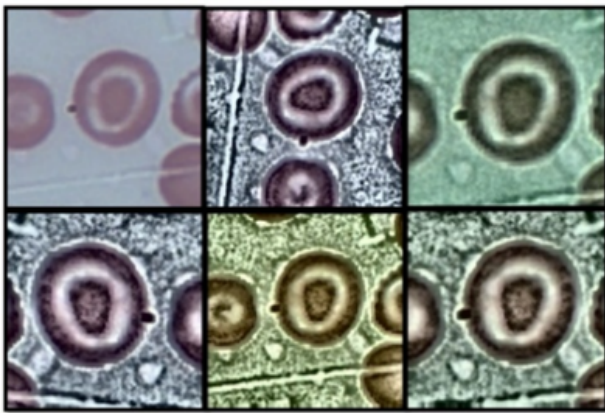


Figure 5: Image Augmentation. (Top left) Original Image. (Top center to Bottom Left) Consecutive multiple augmentations on the origin image.

III. DESIGN AND TRAINING OF NEW FRAMEWORK

The design of the new AI-framework also comprised the use of the U-net architecture and the autoencoder and transfer learning.

A. The U-net Architecture

U-net which is one of the most applied architectures for semantic image segmentation especially in the medical field, was implemented. It consisted of two parts: an encoder to the left, that captures context, and a decoder to the right, that aids in precise localization, forming a U-shape hence its name (see Fig. 6). At a lower level, the neurons learnt features for a small region of the image, such as edges and colors. Whereas at a higher level the neurons learnt features for a large region of the image, such as different objects. Thus, as the number of layers increased the image size decreased, and the number of channels kept on increasing. The down sampling was done by pooling layers. Whereas up sampling was done by transposed convolution. To retain spatial information, a U-net used only convolutional layers and avoided fully connected layers. The convolutional layers coupled with down sampling layers produced a low-resolution image containing the high level of information.

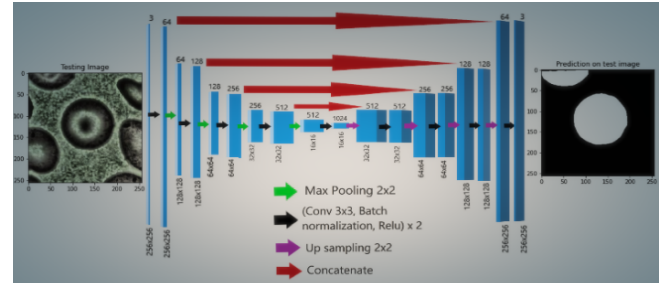


Figure 6: U-net architecture with prediction scenario.

The convolutional layers coupled with up sampling layers produced a high-resolution segmentation output with low level of information. Loss of low-level information led to inaccurate predicted boundaries in segmentation maps produced by the decoder. To counter that skip connections were used to allow the decoder access low-level features produced by the encoder layers. Intermediate outputs from the encoder layers were added with the inputs of the intermediate layers of the decoder at certain positions, producing accurate boundaries.

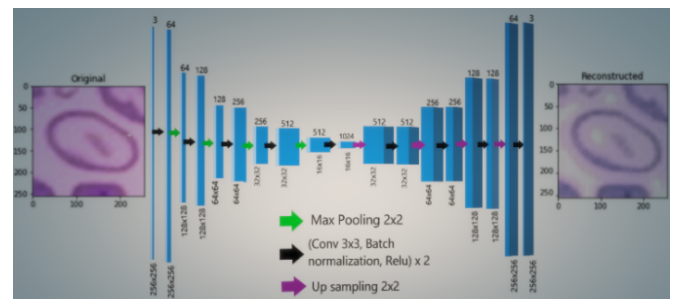


Figure 7: Auto encoder architecture with prediction scenario.

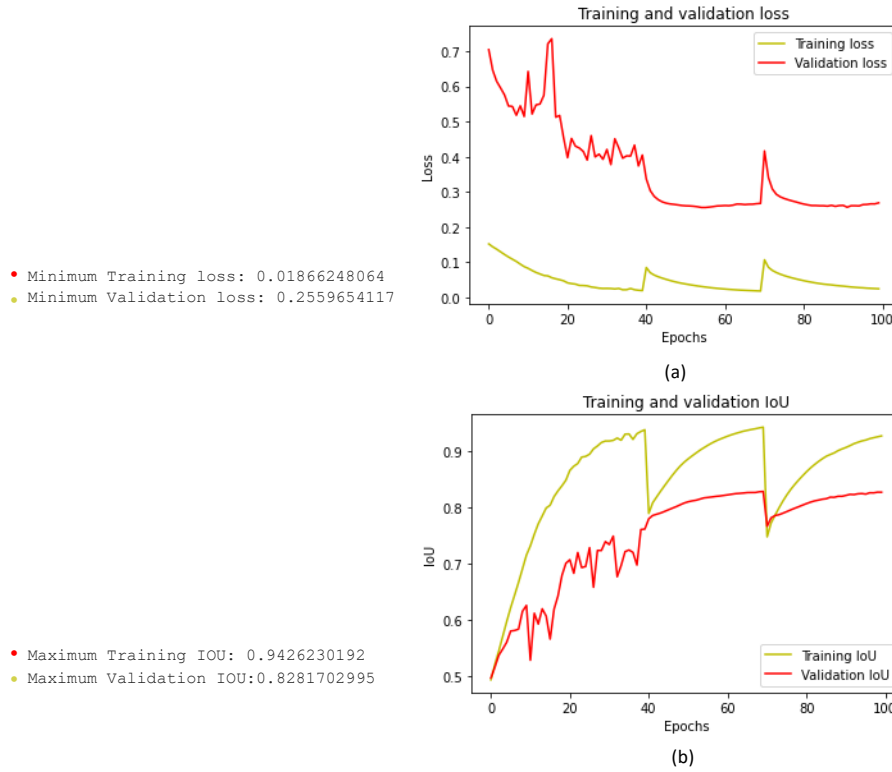


Figure 9: AI-Based Framework Results for Detection and Diagnosis of Thalassemia. (a) Plot of Training (annotated in light green) and Validation (annotated in red) of the new combined metric of loss core versus the epochs with the minimum training and validation loss score values reported. (b) Plot of Training (annotated in light green) and Validation (annotated in red) of the IoU metric versus the epochs with the maximum training and validation IoU metric reported.

B. The Auto Encoder and Transfer Learning

The Auto Encoder follows an architecture similar to that of U-net, it was designed in a way such that its encoder part was identical to that of U-net, and hence its encoder weights could be utilized as a backbone for the U-net model. Transfer Learning was used to make up for the lack of computational power and data resources, speeding up both the training time and converging rate (see Fig. 7). However, most medical image classification studies that employ transfer learning from pretrained models, such as ImageNet, has been proven to be ineffective due to the mismatch in learnt features between the medical images and the natural images provided by ImageNet. To overcome this issue, an autoencoder model was trained on large unlabeled augmented Codocyte images. And once the autoencoder outputs a good prediction, the resulting weights of the encoder part of the autoencoder were transferred to the encoder part of the U-net model as initialized weights, which was later trained on labeled data. The U-net model was trained with frozen backbone parameters for 50 epochs and then for another 100 epochs with unfrozen layers. Then evaluation was applied through the calculation of metrics.

IV. METRICS AND LOSS FUNCTIONS

The metrics utilized in the new AI-based framework were the Jaccard Loss or Intersection over Union (IoU), and the new combined metric of loss scores in order to evaluate the new AI-based framework in detecting and diagnosing Thalassemia.

A. Jaccard Loss (IoU)

The Jaccard Loss also known as IoU is used to quantify the overlapping percentage between the ground truth and the output prediction as shown in Fig. 8. The IoU is given as follows:

$$IoU = \frac{TP}{(TP+FP+FN)} \quad (1)$$

where:

- TP is the True Positive
- FP is the False Positive
- FN is the False Negative.

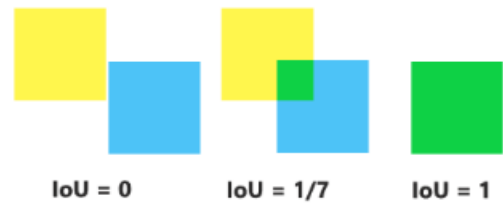


Figure 8: IoU Visualization.

B. Loss Functions: Binary Focal Jaccard Loss

Binary Focal Jaccard Loss is used in the U-net model. It is a combination between the well-known Binary Focal Loss and Jaccard Loss and it is given as follows:

$$\text{Binary Focal Jaccard Loss} = \text{Binary Focal} + \text{Jaccard} \quad (2)$$

a) Binary Focal Loss

This loss function generalizes binary cross-entropy by introducing a hyperparameter γ that gives more weight for the hard, misclassified, false negative samples; than easy, correctly classified, true negative samples and it is given as follows:

$$FL = -\alpha y(1-p)^\gamma \log(p) - (1-y)p^\gamma \log(1-p) \quad (3)$$

where in equation (3):

- $y \in \{0,1\}$ is a binary class label
- $p \in [0,1]$ is a probability of the positive class
- γ specifies how much higher-confidence correct predictions contribute to the overall loss
- α is a hyperparameter that governs the trade-off between precision and recall by weighting errors for the positive class.

V. RESULTS

The new AI-based Framework results were shown through the U-net graph plots of the new metric of combined loss scores and IoU shown in Figs. 9 (a) and (b), and the PTA shown in Figs. 10 (a) and (b), respectively.

A. U-net Graph Plots of Loss Score and IoU

At the end of training, the loss score was approximately 0.01866 and the IoU score was 0.94262. While at the end of validation, the loss score was 0.25596, and the IoU score was 0.82817. In Fig. 9, the sudden drops in the score occurring in the graphs was due to re-initializing the training set with data augmentation each 40 epochs or so. Quantitative results in percentage showed that, the mean IoU score of prediction of Thalassemia was 88% with Prediction Time Augmentation (PTA) and 82% without PTA. Results also showed that as the combined metric of loss scores decreases the prediction of Thalassemia increases. The results were improved by using PTA.

B. Prediction Time Augmentation (PTA)

PTA was implemented to make up for any human-error done during annotation and help optimize the prediction minimizing mislabeled regions as much as possible as in the study of Moshkov et al. [9]. Augmentation was done on the input image, where it was flipped and rotated in all directions. Then the U-net model generated a prediction for each of the augmented images. Finally, all the generated predictions were averaged out and compared to a threshold to form the final optimized smoothed prediction of Thalassemia. The qualitative results were shown in Fig. 10, the final prediction of Thalassemia focused on the Codocytes and labeled the other unidentified cells as the background. Also, the resulting image was smoother and less bulky than the original annotated ground truth, and thus could be feasibly diagnosed.

VI. DISCUSSION AND CONCLUSION

The results of PTA, loss scores and IoU of the new AI-Based Framework were promising in providing an automatic and improved diagnosis of Thalassemia. The evaluation of

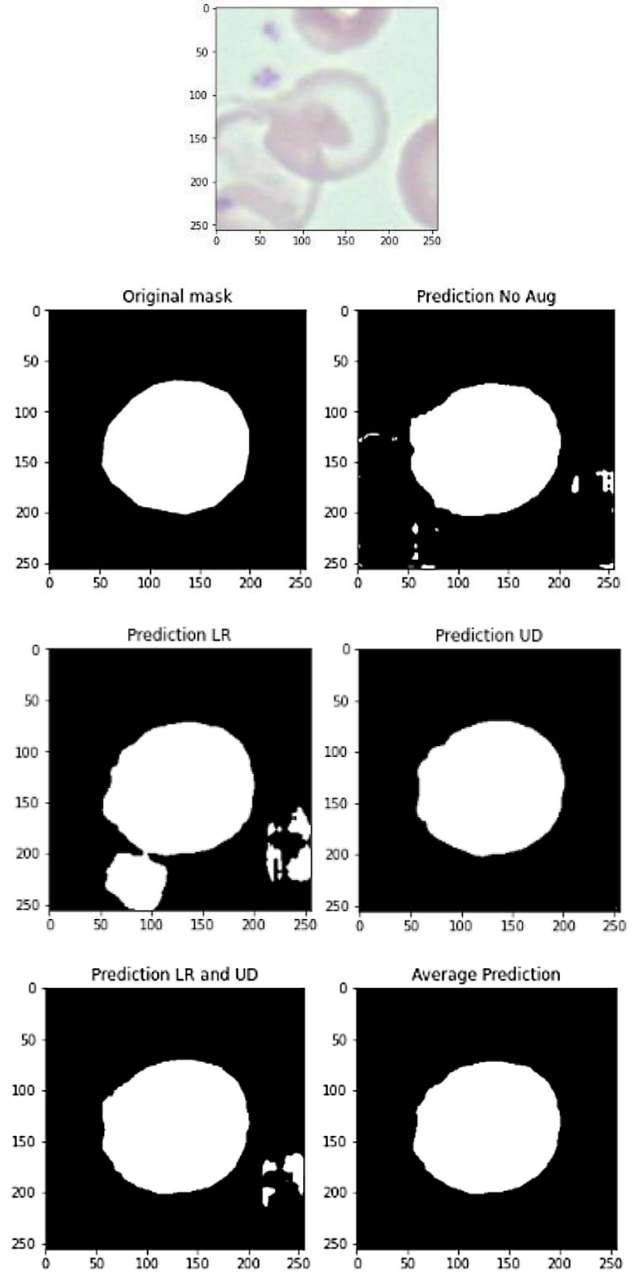


Figure 10: Qualitative Results of Prediction Time Augmentation (PTA) Visualization. Starting from predicting the mask when: no augmentation exists, when rotating exists from Left to Right (LR), when flipping exists from Up to Down (UD), when rotation LR and flipping UD exist, ending with the Average prediction image of all cases.

the new metric of the combination of loss scores provided in this work was used for the first time and was not evaluated in other research papers discussed in the literature. Although some attempts employed the use of AI, however, the nets used in the current work were based on DL rather than ML as compared to existing papers [4,5].

Quantitative results showed that, the mean IoU score of prediction of Thalassemia in percentage was 88% with PTA and 82% without PTA. Results also showed that as the new combined metric of loss scores decreases the prediction of

Thalassemia increases. Qualitative Results showed that the final prediction of Thalassemia focused on Codocytes and labeled the other unidentified cells as the background. Also, it is smoother and less bulky than the original annotated ground truth, and thus could be feasibly diagnosed.

The current work summarizes the whole process pipeline of designing and developing the new AI-based framework for Thalassemia diagnosis. The data collection and preparation, as well as data annotation and augmentation were described. The way transfer learning was achieved and the architecture of the U-net and Autoencoder models were explained. The new metrics of loss scores and IoU used by each model were provided and PTA was implemented. Finally, several future work goals and briefly highlights on how to proceed on applying them.

VII. FUTURE WORK

Some of the future goals include training the model on a larger dataset to achieve better convergence, devise a way to perform predictions on large sized images by developing small tiled overlapping patches that fit the current model's input image dimensions, run the prediction on them, then merge the output to get the final prediction. Another goal is to extract information from each ROI, such as its equivalent area, etc. and to extend the model to perform multiclass image segmentation on other abnormal red blood cell morphologies.

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